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# SECURE ENTERPRISE NETWORKING

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AUGUST 7, 2025  
AYDEN ROMULUS

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## Part A – The Diagram

### Introduction

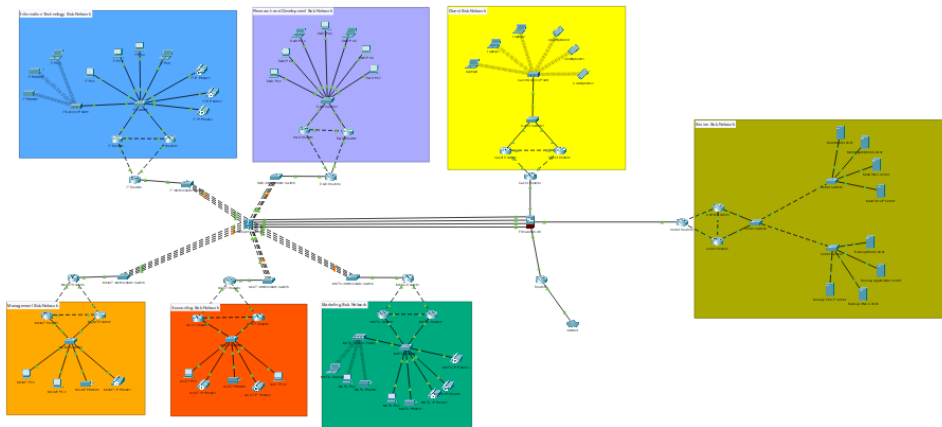
This section of the document shows the design of a secure and resilient network for a small start-up company which adopts realistic concepts utilized by the majority of organizations today. This company will have the following departments: Management, Finance, Marketing, IT, R & D, which will all have their own dedicated sub networks. There is also an additional sub network for Guests which will be separated accordingly.

Since this company is regarded as a start-up, it is highly likely to expand in the future. To deal with this possibility, the network has been designed with scalability in mind. Moreover, the network has also been designed to address and mitigate the impact of a single point of failure (SPOF) as much as possible.

The network will also include internal servers like email, application, Domain Name Service (DNS), and Dynamic Host Configuration Protocol (DHCP). The network will also display a connection to the internet as well as employ security features like a firewall to protect the operations of the internal network.

## Implementation of Network

Below is a screenshot of the network in its entirety.



In the image, the colored squares show all the different sub networks and how each of them is connected to each other. In the next few sub chapters, the design features of the network will be discussed in greater detail.

## Logical Segmentation of Departments

All departments in the network have been logically separated or segmented showing traffic isolation through utilizing Virtual Local Area Networks (VLANs) and subnets. For example:

Figure 1  
Topology of the IT department

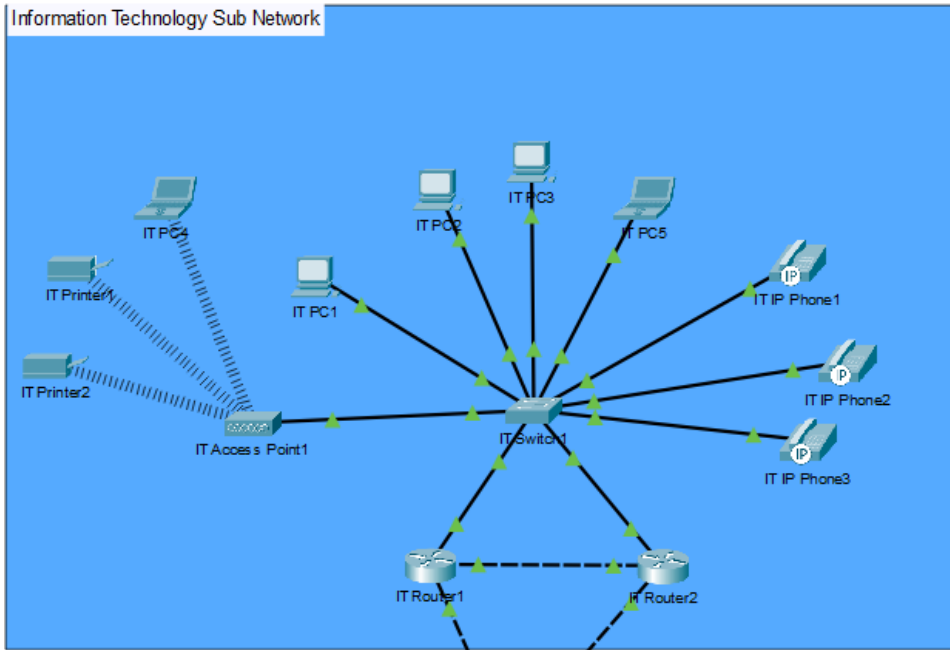


Figure 2  
VLAN configuration of the IT department

| Device Name: IT Switch1         |      |      |                |                |
|---------------------------------|------|------|----------------|----------------|
| Custom Device Model: 2960 IOS15 |      |      |                |                |
| Hostname: IT-Switch1            |      |      |                |                |
| Port                            | Link | VLAN | IP Address     | MAC Address    |
| FastEthernet0/1                 | Up   | 10   | --             | 0060.471D.E601 |
| FastEthernet0/2                 | Up   | 10   | --             | 0060.471D.E602 |
| FastEthernet0/3                 | Up   | 10   | --             | 0060.471D.E603 |
| FastEthernet0/4                 | Up   | 10   | --             | 0060.471D.E604 |
| FastEthernet0/5                 | Up   | 10   | --             | 0060.471D.E605 |
| FastEthernet0/6                 | Up   | 10   | --             | 0060.471D.E606 |
| FastEthernet0/7                 | Up   | 10   | --             | 0060.471D.E607 |
| FastEthernet0/8                 | Up   | 10   | --             | 0060.471D.E608 |
| FastEthernet0/9                 | Down | 1    | --             | 0060.471D.E609 |
| FastEthernet0/10                | Down | 1    | --             | 0060.471D.E60A |
| FastEthernet0/11                | Down | 1    | --             | 0060.471D.E60B |
| FastEthernet0/12                | Down | 1    | --             | 0060.471D.E60C |
| FastEthernet0/13                | Down | 1    | --             | 0060.471D.E60D |
| FastEthernet0/14                | Down | 1    | --             | 0060.471D.E60E |
| FastEthernet0/15                | Down | 1    | --             | 0060.471D.E60F |
| FastEthernet0/16                | Down | 1    | --             | 0060.471D.E610 |
| FastEthernet0/17                | Down | 1    | --             | 0060.471D.E611 |
| FastEthernet0/18                | Down | 1    | --             | 0060.471D.E612 |
| FastEthernet0/19                | Down | 1    | --             | 0060.471D.E613 |
| FastEthernet0/20                | Down | 1    | --             | 0060.471D.E614 |
| FastEthernet0/21                | Down | 1    | --             | 0060.471D.E615 |
| FastEthernet0/22                | Down | 1    | --             | 0060.471D.E616 |
| FastEthernet0/23                | Down | 1    | --             | 0060.471D.E617 |
| FastEthernet0/24                | Down | 1    | --             | 0060.471D.E618 |
| GigabitEthernet0/1              | Up   | 1    | --             | 0060.471D.E619 |
| GigabitEthernet0/2              | Up   | 1    | --             | 0060.471D.E61A |
| Vlan1                           | Down | 1    | <not set>      | 0001.431D.836D |
| Vlan10                          | Up   | 10   | 172.16.10.3/24 | 0001.431D.8301 |

Physical Location: Intercity > Home City > Corporate Office > IT Wiring Closet > Rack > IT Switch1

Figure 3

Topology of the R&D department

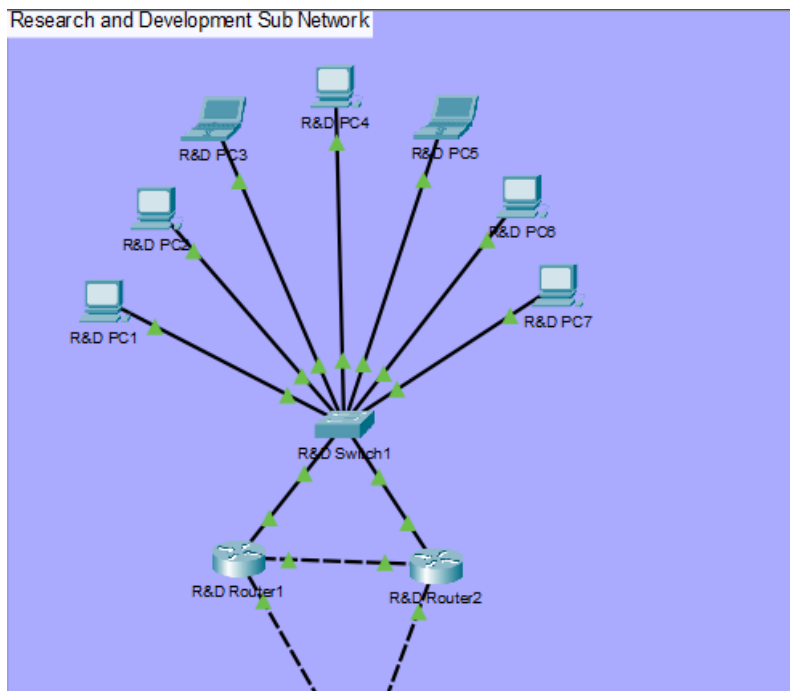


Figure 4

VLAN configuration of the R&D department

| Device Name: R&D Switch1        |      |      |                |                |
|---------------------------------|------|------|----------------|----------------|
| Custom Device Model: 2960 IOS15 |      |      |                |                |
| Hostname: R&D-Switch1           |      |      |                |                |
| Port                            | Link | VLAN | IP Address     | MAC Address    |
| FastEthernet0/1                 | Up   | 20   | --             | 0001.C94E.8701 |
| FastEthernet0/2                 | Up   | 20   | --             | 0001.C94E.8702 |
| FastEthernet0/3                 | Up   | 20   | --             | 0001.C94E.8703 |
| FastEthernet0/4                 | Up   | 20   | --             | 0001.C94E.8704 |
| FastEthernet0/5                 | Up   | 20   | --             | 0001.C94E.8705 |
| FastEthernet0/6                 | Up   | 20   | --             | 0001.C94E.8706 |
| FastEthernet0/7                 | Up   | 20   | --             | 0001.C94E.8707 |
| FastEthernet0/8                 | Down | 1    | --             | 0001.C94E.8708 |
| FastEthernet0/9                 | Down | 1    | --             | 0001.C94E.8709 |
| FastEthernet0/10                | Down | 1    | --             | 0001.C94E.870A |
| FastEthernet0/11                | Down | 1    | --             | 0001.C94E.870B |
| FastEthernet0/12                | Down | 1    | --             | 0001.C94E.870C |
| FastEthernet0/13                | Down | 1    | --             | 0001.C94E.870D |
| FastEthernet0/14                | Down | 1    | --             | 0001.C94E.870E |
| FastEthernet0/15                | Down | 1    | --             | 0001.C94E.870F |
| FastEthernet0/16                | Down | 1    | --             | 0001.C94E.8710 |
| FastEthernet0/17                | Down | 1    | --             | 0001.C94E.8711 |
| FastEthernet0/18                | Down | 1    | --             | 0001.C94E.8712 |
| FastEthernet0/19                | Down | 1    | --             | 0001.C94E.8713 |
| FastEthernet0/20                | Down | 1    | --             | 0001.C94E.8714 |
| FastEthernet0/21                | Down | 1    | --             | 0001.C94E.8715 |
| FastEthernet0/22                | Down | 1    | --             | 0001.C94E.8716 |
| FastEthernet0/23                | Down | 1    | --             | 0001.C94E.8717 |
| FastEthernet0/24                | Down | 1    | --             | 0001.C94E.8718 |
| GigabitEthernet0/1              | Up   | 1    | --             | 0001.C94E.8719 |
| GigabitEthernet0/2              | Up   | 1    | --             | 0001.C94E.871A |
| Vlan1                           | Down | 1    | <not set>      | 0001.C713.E825 |
| Vlan20                          | Up   | 20   | 172.16.20.3/24 | 0001.C713.E801 |

Physical Location: Intercity > Home City > Corporate Office > R&D Wiring Closet > Rack > R&D Switch1

Figure 5

Topology of the Management, Accounting and Marketing department

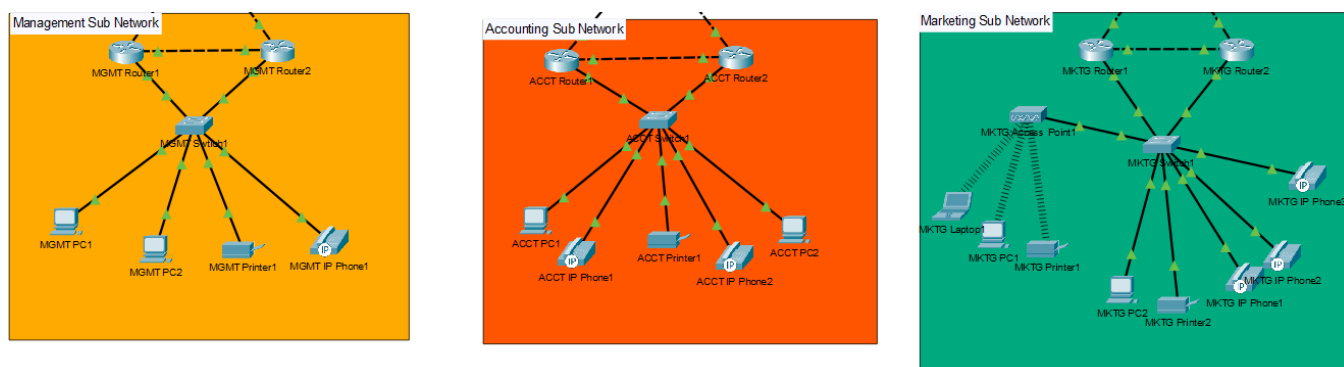


Figure 6

VLAN configuration of the Management department

| Device Name: MGMT Swtich1 |      |      |                |                |
|---------------------------|------|------|----------------|----------------|
| Device Model: 2960-24TT   |      |      |                |                |
| Hostname: ACCT-Switch1    |      |      |                |                |
| Port                      | Link | VLAN | IP Address     | MAC Address    |
| FastEthernet0/1           | Up   | 30   | --             | 0040.0BE3.3201 |
| FastEthernet0/2           | Up   | 30   | --             | 0040.0BE3.3202 |
| FastEthernet0/3           | Up   | 30   | --             | 0040.0BE3.3203 |
| FastEthernet0/4           | Up   | 30   | --             | 0040.0BE3.3204 |
| FastEthernet0/5           | Down | 1    | --             | 0040.0BE3.3205 |
| FastEthernet0/6           | Down | 1    | --             | 0040.0BE3.3206 |
| FastEthernet0/7           | Down | 1    | --             | 0040.0BE3.3207 |
| FastEthernet0/8           | Down | 1    | --             | 0040.0BE3.3208 |
| FastEthernet0/9           | Down | 1    | --             | 0040.0BE3.3209 |
| FastEthernet0/10          | Down | 1    | --             | 0040.0BE3.320A |
| FastEthernet0/11          | Down | 1    | --             | 0040.0BE3.320B |
| FastEthernet0/12          | Down | 1    | --             | 0040.0BE3.320C |
| FastEthernet0/13          | Down | 1    | --             | 0040.0BE3.320D |
| FastEthernet0/14          | Down | 1    | --             | 0040.0BE3.320E |
| FastEthernet0/15          | Down | 1    | --             | 0040.0BE3.320F |
| FastEthernet0/16          | Down | 1    | --             | 0040.0BE3.3210 |
| FastEthernet0/17          | Down | 1    | --             | 0040.0BE3.3211 |
| FastEthernet0/18          | Down | 1    | --             | 0040.0BE3.3212 |
| FastEthernet0/19          | Down | 1    | --             | 0040.0BE3.3213 |
| FastEthernet0/20          | Down | 1    | --             | 0040.0BE3.3214 |
| FastEthernet0/21          | Down | 1    | --             | 0040.0BE3.3215 |
| FastEthernet0/22          | Down | 1    | --             | 0040.0BE3.3216 |
| FastEthernet0/23          | Down | 1    | --             | 0040.0BE3.3217 |
| FastEthernet0/24          | Down | 1    | --             | 0040.0BE3.3218 |
| GigabitEthernet0/1        | Up   | 1    | --             | 0040.0BE3.3219 |
| GigabitEthernet0/2        | Up   | 1    | --             | 0040.0BE3.321A |
| Vlan1                     | Down | 1    | 172.16.30.3/24 | 0001.6441.5DC8 |
| Vlan30                    | Up   | 30   | 172.16.30.3/24 | 0001.6441.5D01 |

Physical Location: Intercity > Home City > Corporate Office > MGMT Wiring Closet > Rack > MGMT Swtich1



Figure 7

## VLAN configuration of Accounting department

| Device Name: ACCT Switch1                                                                                    |      |      |                |                |
|--------------------------------------------------------------------------------------------------------------|------|------|----------------|----------------|
| Custom Device Model: 2960 IOS15                                                                              |      |      |                |                |
| Hostname: ACCT-Switch1                                                                                       |      |      |                |                |
| Port                                                                                                         | Link | VLAN | IP Address     | MAC Address    |
| FastEthernet0/1                                                                                              | Up   | 40   | --             | 000C.85B4.9D01 |
| FastEthernet0/2                                                                                              | Up   | 40   | --             | 000C.85B4.9D02 |
| FastEthernet0/3                                                                                              | Up   | 40   | --             | 000C.85B4.9D03 |
| FastEthernet0/4                                                                                              | Up   | 40   | --             | 000C.85B4.9D04 |
| FastEthernet0/5                                                                                              | Up   | 40   | --             | 000C.85B4.9D05 |
| FastEthernet0/6                                                                                              | Down | 1    | --             | 000C.85B4.9D06 |
| FastEthernet0/7                                                                                              | Down | 1    | --             | 000C.85B4.9D07 |
| FastEthernet0/8                                                                                              | Down | 1    | --             | 000C.85B4.9D08 |
| FastEthernet0/9                                                                                              | Down | 1    | --             | 000C.85B4.9D09 |
| FastEthernet0/10                                                                                             | Down | 1    | --             | 000C.85B4.9D0A |
| FastEthernet0/11                                                                                             | Down | 1    | --             | 000C.85B4.9D0B |
| FastEthernet0/12                                                                                             | Down | 1    | --             | 000C.85B4.9D0C |
| FastEthernet0/13                                                                                             | Down | 1    | --             | 000C.85B4.9D0D |
| FastEthernet0/14                                                                                             | Down | 1    | --             | 000C.85B4.9D0E |
| FastEthernet0/15                                                                                             | Down | 1    | --             | 000C.85B4.9D0F |
| FastEthernet0/16                                                                                             | Down | 1    | --             | 000C.85B4.9D10 |
| FastEthernet0/17                                                                                             | Down | 1    | --             | 000C.85B4.9D11 |
| FastEthernet0/18                                                                                             | Down | 1    | --             | 000C.85B4.9D12 |
| FastEthernet0/19                                                                                             | Down | 1    | --             | 000C.85B4.9D13 |
| FastEthernet0/20                                                                                             | Down | 1    | --             | 000C.85B4.9D14 |
| FastEthernet0/21                                                                                             | Down | 1    | --             | 000C.85B4.9D15 |
| FastEthernet0/22                                                                                             | Down | 1    | --             | 000C.85B4.9D16 |
| FastEthernet0/23                                                                                             | Down | 1    | --             | 000C.85B4.9D17 |
| FastEthernet0/24                                                                                             | Down | 1    | --             | 000C.85B4.9D18 |
| GigabitEthernet0/1                                                                                           | Up   | 1    | --             | 000C.85B4.9D19 |
| GigabitEthernet0/2                                                                                           | Up   | 1    | --             | 000C.85B4.9D1A |
| Vlan1                                                                                                        | Down | 1    | <not set>      | 0009.7C37.99A1 |
| Vlan40                                                                                                       | Up   | 40   | 172.16.40.3/24 | 0009.7C37.9901 |
| Physical Location: Intercity > Home City > Corporate Office > Accounting Wiring Closet > Rack > ACCT Switch1 |      |      |                |                |

Figure 8

## VLAN configuration of Marketing department

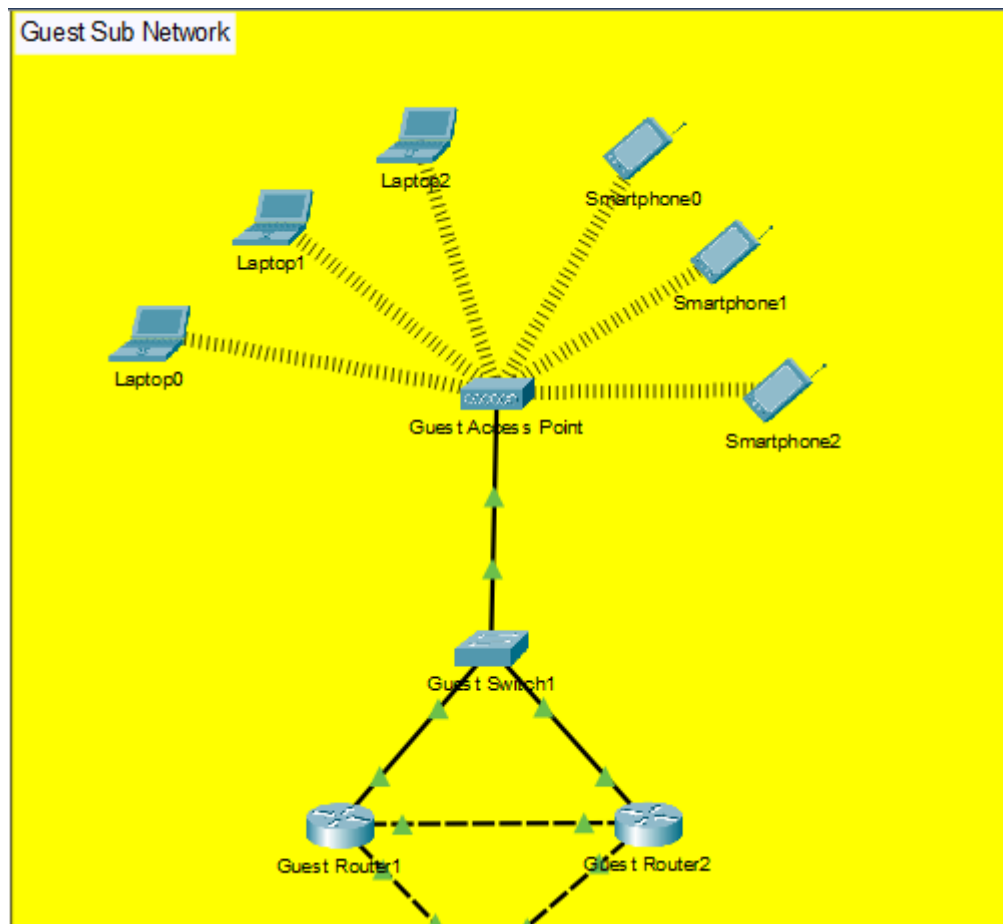
| Device Name: MKTG Switch1                                                                                   |      |      |                |                |
|-------------------------------------------------------------------------------------------------------------|------|------|----------------|----------------|
| Custom Device Model: 2960 IOS15                                                                             |      |      |                |                |
| Hostname: MKTG-Switch1                                                                                      |      |      |                |                |
| Port                                                                                                        | Link | VLAN | IP Address     | MAC Address    |
| FastEthernet0/1                                                                                             | Up   | 50   | --             | 0030.A3BE.A901 |
| FastEthernet0/2                                                                                             | Up   | 50   | --             | 0030.A3BE.A902 |
| FastEthernet0/3                                                                                             | Up   | 50   | --             | 0030.A3BE.A903 |
| FastEthernet0/4                                                                                             | Up   | 50   | --             | 0030.A3BE.A904 |
| FastEthernet0/5                                                                                             | Up   | 50   | --             | 0030.A3BE.A905 |
| FastEthernet0/6                                                                                             | Up   | 50   | --             | 0030.A3BE.A906 |
| FastEthernet0/7                                                                                             | Down | 1    | --             | 0030.A3BE.A907 |
| FastEthernet0/8                                                                                             | Down | 1    | --             | 0030.A3BE.A908 |
| FastEthernet0/9                                                                                             | Down | 1    | --             | 0030.A3BE.A909 |
| FastEthernet0/10                                                                                            | Down | 1    | --             | 0030.A3BE.A90A |
| FastEthernet0/11                                                                                            | Down | 1    | --             | 0030.A3BE.A90B |
| FastEthernet0/12                                                                                            | Down | 1    | --             | 0030.A3BE.A90C |
| FastEthernet0/13                                                                                            | Down | 1    | --             | 0030.A3BE.A90D |
| FastEthernet0/14                                                                                            | Down | 1    | --             | 0030.A3BE.A90E |
| FastEthernet0/15                                                                                            | Down | 1    | --             | 0030.A3BE.A90F |
| FastEthernet0/16                                                                                            | Down | 1    | --             | 0030.A3BE.A910 |
| FastEthernet0/17                                                                                            | Down | 1    | --             | 0030.A3BE.A911 |
| FastEthernet0/18                                                                                            | Down | 1    | --             | 0030.A3BE.A912 |
| FastEthernet0/19                                                                                            | Down | 1    | --             | 0030.A3BE.A913 |
| FastEthernet0/20                                                                                            | Down | 1    | --             | 0030.A3BE.A914 |
| FastEthernet0/21                                                                                            | Down | 1    | --             | 0030.A3BE.A915 |
| FastEthernet0/22                                                                                            | Down | 1    | --             | 0030.A3BE.A916 |
| FastEthernet0/23                                                                                            | Down | 1    | --             | 0030.A3BE.A917 |
| FastEthernet0/24                                                                                            | Down | 1    | --             | 0030.A3BE.A918 |
| GigabitEthernet0/1                                                                                          | Up   | 1    | --             | 0030.A3BE.A919 |
| GigabitEthernet0/2                                                                                          | Up   | 1    | --             | 0030.A3BE.A91A |
| Vlan1                                                                                                       | Down | 1    | <not set>      | 0009.7CE7.9710 |
| Vlan50                                                                                                      | Up   | 50   | 172.16.50.3/24 | 0009.7CE7.9701 |
| Physical Location: Intercity > Home City > Corporate Office > Marketing Wiring Closet > Rack > MKTG Switch1 |      |      |                |                |

## Device Allocation: Printers, IP Phones

All departments contain printers and IP phones except for the R&D sub network as well as the guest network. Regarding the R&D department, this is most likely done because of privacy concerns ([See Fig. 3](#)). The guest network's purpose is to provide external users with internet access only. External users are not required to have access to IP phones or printers.

*Figure 9*

Topology of the guest network



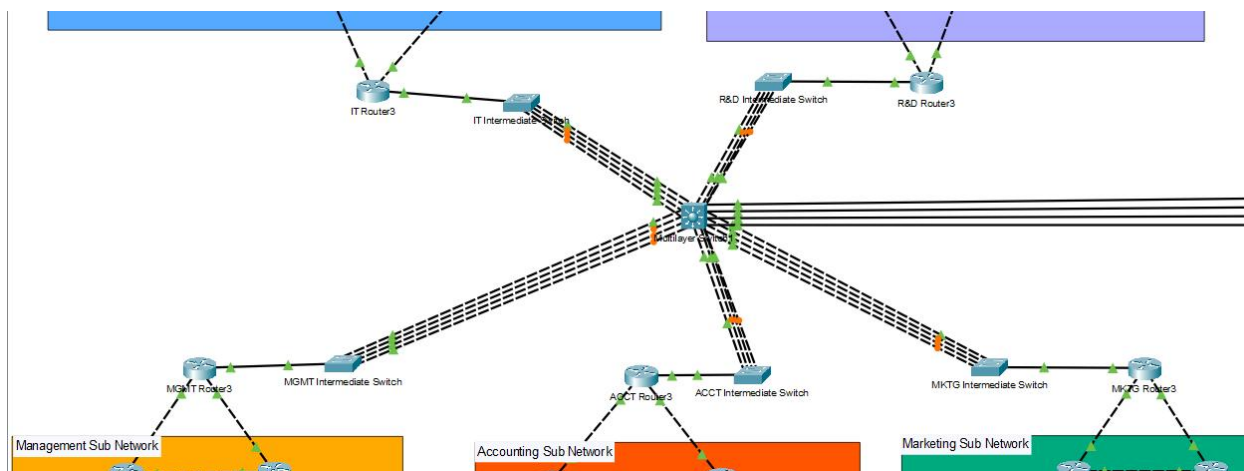
## Scalability and Resilience/Redundancy

The network is designed using a modular hierarchical architecture that separates functionality into core distribution and access layers. This structure promotes scalability, reliability, and simplified management. Each department operates within its own isolated subnet with dedicated and backup routers as well as intermediate switches. The network also shows unique VLANs and IP addressing schemes for hosts. This potentially allows for additional departments or units to be integrated seamlessly by replicating the existing structure.

The network also supports a centralized core using a multilayer switch. Departments are connected through EtherChannel or utilizing multiple links to the multilayer switch to reduce SPOF.

*Figure 10*

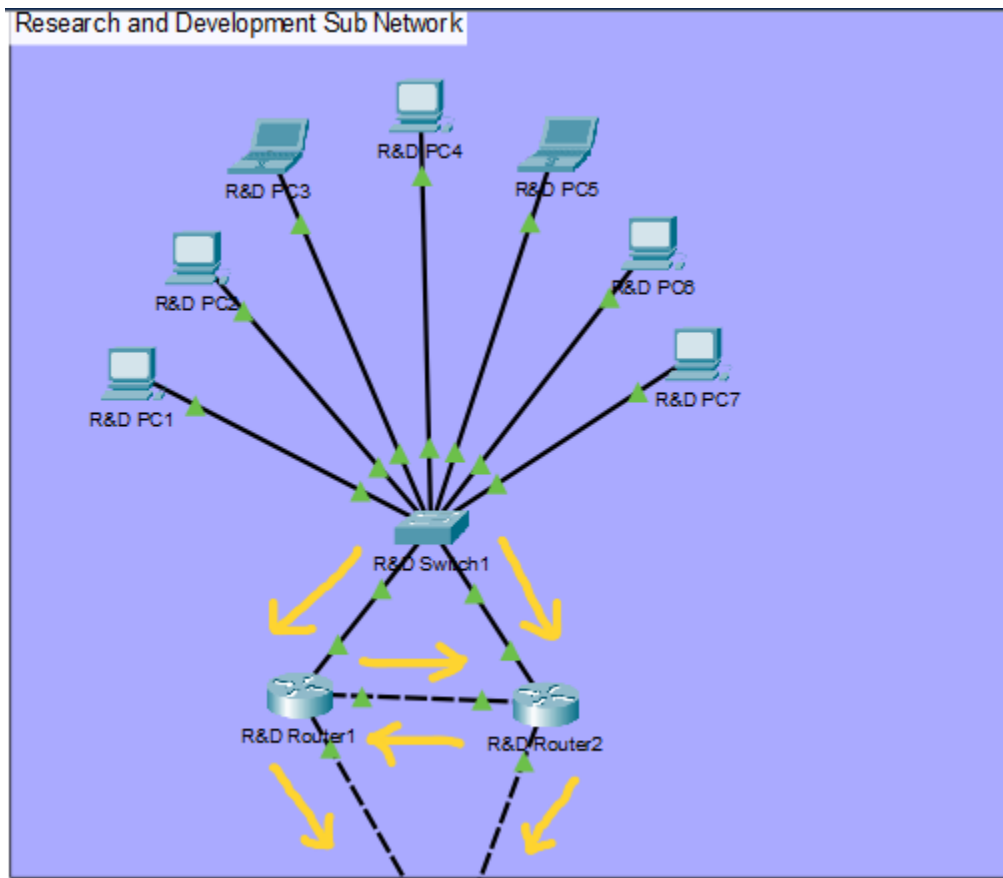
All departments using multiple links to the multilayer switch.



Internally, it uses methods like Open Shortest Path First (OSPF) to dynamically route traffic if a router were to go offline. This means new connections can be added without modifying existing paths.

*Figure 11*

Illustration of OSPF



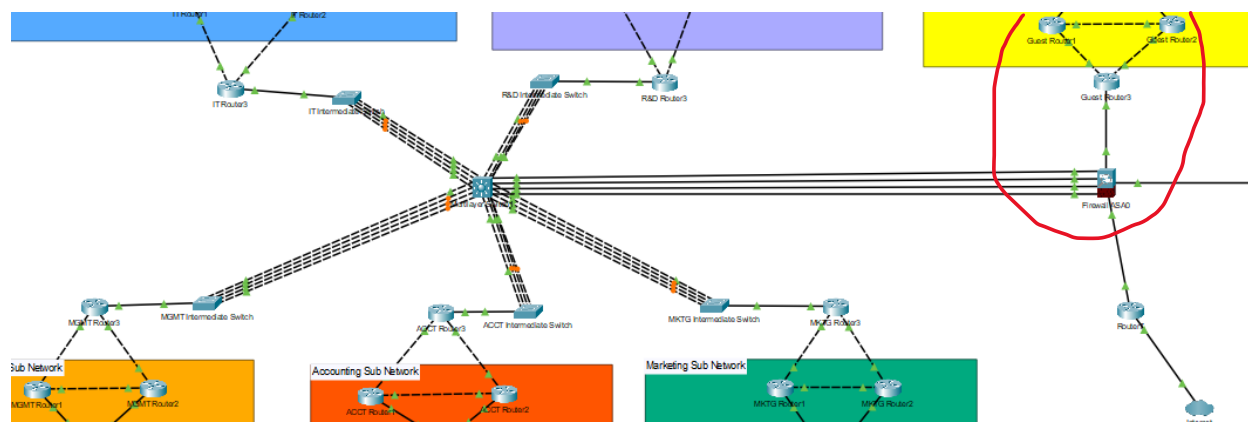
The arrows signify the paths traffic can take.

## Secure Wireless Access Control

The guest network is entirely isolated from the department networks within the company. This is done by using a firewall.

*Figure 12*

Guest network isolated from the rest via a firewall.



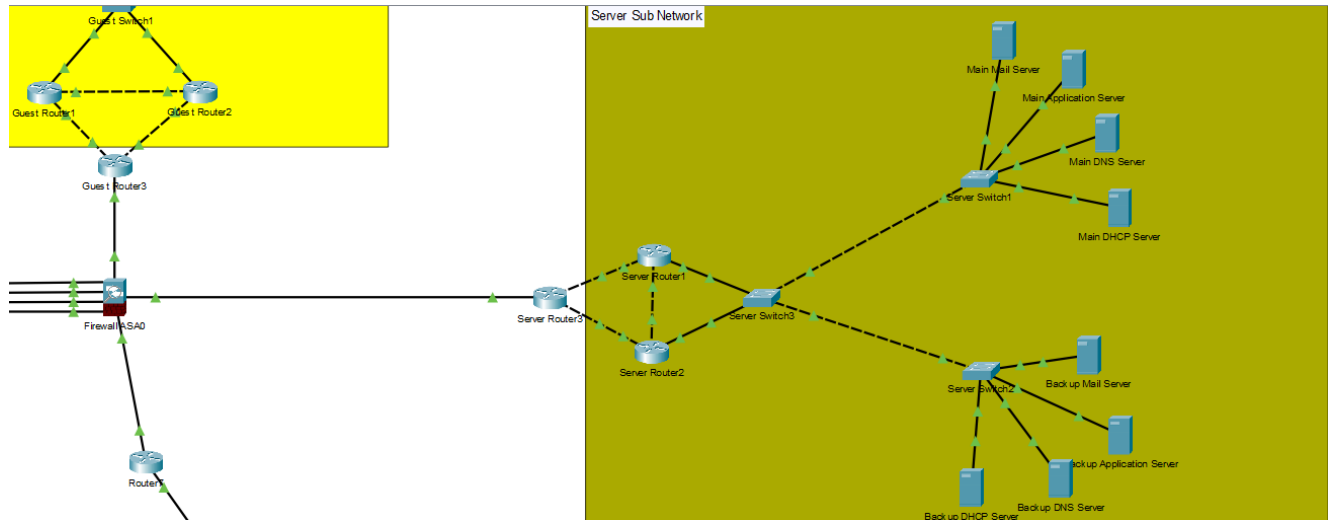
The red markup shows the guest network being connected to the firewall.

## Server Placement

All servers (mail, application, DNS, DHCP) have been placed appropriately in a Demilitarized Zone (DMZ). DMZ is done by segregating devices on the opposite side of a firewall.

*Figure 13*

Main and Backup Servers are placed in DMZ (placed on the opposite side of the firewall) for security reasons.

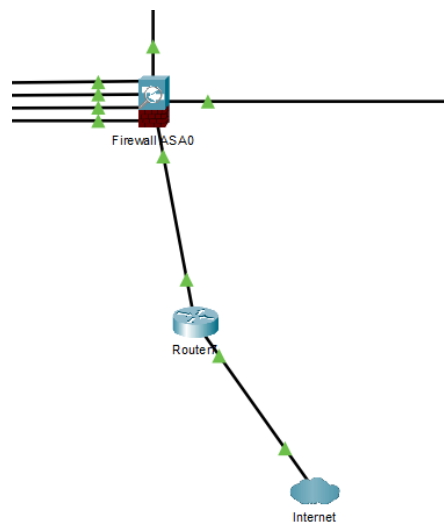


## Internet Connection

In the network topology, all sub networks have access to the internet. The internet has been securely segmented from and connected to the internal network via the firewall.

*Figure 14*

Internet connected to the internal network through the firewall.



## Part B – The Theory

What is the difference, if any, between a layer two and a layer three switch?

A layer two switch is a network device that operates at the data link layer (Layer 2) of the OSI model. Its primary function is to forward data frames based on MAC addresses and Ethernet frames, effectively connecting devices within a local area network (LAN).

A layer three switch is a network device that combines the switching functions of a Layer 2 switch with the routing capabilities of a router. A layer three switch can forward traffic within a LAN based on MAC addresses as well as route traffic between networks based on IP addresses. It could be said that a layer three switch is smarter than a layer two switch.

Below is a table of their brief differences:

| Layer 2 Switch                         | Layer 3 Switch                          |
|----------------------------------------|-----------------------------------------|
| OSI Layer 2 (Data Link)                | OSI Layer 2 & 3 (Data Link and Network) |
| Communicates using MAC Addresses       | Communicates using Mac and IP Addresses |
| No Routing Capability                  | Has Routing Capability                  |
| No Inter-VLAN Routing (Needs a Router) | Has Inter-VLAN Routing (Built-in)       |
| Connect devices in the same LAN        | Connect multiple networks at high speed |

Explain how this arrangement reduces the impact of a single point of failure. Give an example showing how the different components add resilience.

Industry best practices were used in the development of this network, which combines modular segmentation, link aggregation (EtherChannel), hierarchical design, and redundancy. Each of these elements ensures that if one device, path or interface were to fail, the network as a whole will not go offline or departments will not lose access to essential services.

### Redundant Routers Per Department

Each department has two routers connected ([see Fig. 5](#)) (e.g., MGMT Router1 and MGMT Router2). These routers are:

- Connected to each other via point-to-point links.
- Both connected through IT Router 3 with the intermediary switch toward the core multilayer switch.
- Runs a dynamic routing protocol (OSPF)

Example:

If IT Router1 fails, all traffic will automatically route through IT Router2, which has its own link towards the core switch and continues using OSPF. Internal devices will remain online without any reconfiguration needed.

### Intermediate Switches with EtherChannel Links

Each department's internal sub network connects to intermediate switches, which establish an EtherChannel (link aggregation control protocol) with the core multilayer switch



([See Fig. 10](#)). EtherChannel allows multiple physical links to function as a single logical link. This allows for increased bandwidth and redundancy.

Example:

If one cable in the EtherChannel fails (e.g., between the MGMT Intermediate Switch and the Core Multilayer Switch), the other links automatically carry the traffic. Users will experience no disruption.

## Hierarchical and Modular Design

The network separates access, distribution, and core layers so that:

- Failure or disruption in one department does not affect the others.
- New departments can be added in the future without redesigning.
- Troubleshooting is localized and efficient.

Example:

If R&D's intermediate switch goes offline, only R&D users are affected. The IT, Management, Accounting, and Marketing subnets will continue operating normally.

## Firewall Positioned for Centralized Control

The firewall is connected to the core multilayer switch, so traffic from any subnet routes through it before reaching the simulated Internet. With EtherChannel, even firewall-facing interfaces gain redundancy ([See Fig. 12](#) and [13](#)).

Example:

If any links from the core to the firewall fail, the other links in the EtherChannel will continue to handle any inbound or outbound traffic, preserving internet access ([See Fig.14](#)).

## Name two departments you would not expect to have a printer or IP phone. Why wouldn't these departments have these devices?

The two departments that are not expected to have printers or IP Phones are the R&D department and the Guest sub network.

The R&D department is a team within a company dedicated to investigating and developing new products, services, and processes. It focuses on innovation and improving existing products which give a competitive edge over competitors in the market.

Anything developed in this department is proprietary and must be kept confidential within the company. Allowing the installation of IP phones can break that confidentiality and important documentation printed from printers if misplaced or stolen can be detrimental.

The Guest sub network would not have access to IP phones and printers because it is configured for external users and is only used for the purpose of accessing the internet.

## Suggest ten firewall rules for the company.

Firewall rules set constraints that a network must follow in order to operate securely. Here are ten firewall rules that would be recommended for this network:

### Deny internet access to printers in all subnets.

Printers should never or be accessible from the public internet as they could be:

- Exploited for remote code execution (RCE)
- Used as points to reach other devices.
- Leaked data sources (e.g., print logs, stored jobs)

Blocking internet access for printers ensures they remain confined within the internal network, reducing both internal and external attack surfaces.

### Allow HTTP/HTTPS from IT, Management, and Marketing subnets to the internet.

This rule allows users in trusted departments to

- Access websites, portals, and application tools
- Perform updates and integrations.
- Conduct legitimate work which requires internet access.

### Deny access from Marketing to R&D subnet.

This is an internal segmentation rule. Even within a trusted network, not all departments should communicate with each other.

- Prevents attack progression and escalation in the event of compromise.
- Ensure that sensitive R&D data stays isolated.
- Enforces business logic (e.g., Marketing does not need access to engineering prototypes or research files)

### Allow DNS queries from all subnets.

This allows internal hosts to send DNS queries typically to external DNS servers or a local forwarder. Without DNS:

- Web browsing, updates, and cloud services will all fail.
- Even allowed traffic (e.g., HTTP) will not work because hosts cannot resolve domain names.

### Deny all traffic from Internet to internal IP phones.

VoIP devices are often vulnerable to:

- Session Initiation Protocol (SIP) brute-force attacks
- Eavesdropping
- Unauthorized call routing or hijacking

Blocking traffic to these devices from the internet hardens the VoIP security of the network, ensuring that only internal, managed traffic can reach them.

### Allow OSPF between internal routers.

This allows internal routers to form OSPF neighbor relationships and exchange routing updates dynamically. OSPF is significant to the design of this network and if routed across the firewall, must be allowed.

### Deny all other inbound traffic from the internet.

This is regarded as a catch-all rule. It

- Ensure that anything not explicitly allowed is denied by default.
- Prevents accidental exposure if a new service is brought online without a rule.

## Justifications for the first three mandatory firewall rules

### Deny Pings from the Internet

Also regarded as ICMP echo requests, this rule blocks pings from the public internet into your internal network. If left unhandled these could be used by threat actors for network reconnaissance to discover active hosts, probe firewall behavior to see which hosts are reachable or amplify DDoS attacks.

Blocking ICMP requests from external sources helps conceal the internal network from scanning tools, reducing the chances of being targeted.

## Deny Mail Leaving the R&D Subnet

This rule prohibits outbound SMTP (email) traffic from hosts within the R&D subnet. Sensitive intellectual property, such as prototypes, code, or proprietary designs, is frequently overseen within R&D environments. There are several concerns associated with granting unrestricted email access like data exfiltration and malware spread.

## Deny Requests from the Internet to R&D

This rule blocks all inbound traffic from the internet to the R&D subnet, ensuring total isolation from external sources. This is crucial for maintaining trade secrets, preventing unauthorized access to critical information, and complying with any security frameworks or industry regulations.

**Name a unique device you would include in your network to enhance security. Justify the inclusion of this device.**

## Intrusion Detection and Prevention System (IDPS)

The current firewall only regulates what traffic enters and exits the network. Combining an intrusion detection system and intrusion prevention system allows them both to share responsibility for identifying, analyzing, and responding to the content of the traffic in real time.

This improves security by identifying malware, suspicious patterns (such as port scanning, brute-force login attempts), policy violations, and zero-day exploits. IPSs automatically blocks or quarantines suspicious traffic before it reaches internal hosts.

Implementing this device makes threat detection proactive because even if traffic passes through the firewall, the IDPS analyzes malicious behavior. It stores logs and alerts for incident response and auditing. It can monitor lateral movement within the network. Lastly, it complements the firewall by reacting to behavior while the firewall enforces rules.

## Subnetting Exercise

Assuming a prefix of /27 on the internal network.

The number of subnets will be:

11111111. 11111111. 11111111.11100000

$2^n \rightarrow 2^3 = 8$  subnets

The number of usable hosts on each subnet will be:

$(2^h) - 2 \rightarrow (2^5) - 2 = 30$  hosts

For each consecutive subnet:

| Network ID    | Usable Range                  | Broadcast IP  |
|---------------|-------------------------------|---------------|
| 172.16.10.0   | 172.16.10.1 – 172.16.10.30    | 172.16.10.31  |
| 172.16.10.32  | 172.16.10.33 – 172.16.10.62   | 172.16.10.63  |
| 172.16.10.64  | 172.16.10.65 – 172.16.10.94   | 172.16.10.95  |
| 172.16.10.96  | 172.16.10.97 – 172.16.10.126  | 172.16.10.127 |
| 172.16.10.128 | 172.16.10.129 – 172.16.10.158 | 172.16.10.159 |
| 172.16.10.160 | 172.16.10.161 – 172.16.10.190 | 172.16.10.191 |
| 172.16.10.192 | 172.16.10.193 – 172.16.10.222 | 172.16.10.223 |
| 172.16.10.224 | 172.16.10.225 – 172.16.10.254 | 172.16.10.255 |

The departments need to be able to communicate with each other across the network, list and justify three strategies you would include to securely facilitate this.

### Use Inter-VLAN Routing on a Multilayer Switch

Allowing hosts from one VLAN to communicate with hosts from another is known as inter-VLAN routing. This can be done by utilizing a multilayer switch by enabling routing between VLANs using Switch Virtual Interfaces (SVIs). To achieve this, IP addresses are assigned to each VLAN interface on the core switch (e.g., VLAN 10, 20, 30 etc.) and the default

gateway of each subnet is set to its corresponding SVI. This strategy is described as a straightforward process to configure, and it allows departments to communicate without complex firewall or router rules.

### Use Static Routes on Department Routers

Static routes on a router are manually configured routes that direct network traffic along a specific path, rather than relying on dynamic routing protocols like OSPF. Since each department router already connects to the core switch, static routes can be added to point to other department networks. This strategy keeps routing simple and predictable and works well in smaller networks allowing for static communication among departments.

### Use a Dedicated “Shared Services” VLAN for Inter-Department Access

A new VLAN could be configured for all departments which host shared services (e.g., an email server or central file server). This strategy can reduce the need for direct communication between departments and also encourage segmentation while still supporting collaboration.

### The switch port connected to the router is going to be of which port type? Why?

The switch port usually used to connect to routers is the Gigabit Ethernet port which supports 1 Gbps (1000 Mbps) transfer speeds. It is typically used for performance, compatibility, and reliability.

### Higher Bandwidth (Speed)

Since this port supports 1000 Mbps or 1 Gbps it is ideally used in networks where multiple devices may be transmitting traffic at once. Departments which use VLANs will need fast internet within their network. Applications which require protocols like VoIP, video, and file

sharing demand higher bandwidth. If a slower port like Fast Ethernet (100 Mbps) is used, it would function as a bottleneck for department traffic.

### Router Interfaces Are Often Gigabit-Only

Many routers like modern Cisco routers only have Gigabit Interfaces. If a switch only supported Fast Ethernet, a speed mismatch would be experienced or may not connect at all without manually adjusting the speed.

### Industry Best Practices

In the industry, professionals design their networks to use higher-speed ports for links from switches to routers, trunk connections between switches and facilitating core switch connections. The port mitigates traffic congestion and can manage more data without any major slowdowns.